

2025

COMPUTER SCIENCE

Paper : CSME-401

(Elective-I : Introduction to Data Science)

Full Marks : 70

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*Answer **question nos. 1, 2, and any four** questions from the rest.1. Answer **any five** questions from the following :

2×5

- ✓(a) What is an unbiased estimator? Give an example of an unbiased estimator.
- (b) Is graphical representation of a normal distribution always symmetrical? Propose your view.
- ✓(c) Is linear regression suitable for classification problems? Justify your answer.
- ✓(d) Explain the significance of p-value in regression analysis.
- ✓(e) Can Decision Tree be used for both continuous and categorical target variables? Justify.
- (f) Is it possible to use K-NN for function approximation problem? Justify.
- ✓(g) A species of animal have normally distributed weights, measured in kilograms, with a mean of 143.5 and a variance of 47.0. The weight of one particular such animal is 159.5 kilogram. Compute the z-score.

2. Answer **any five** questions from the following :

4×5

- ✓(a) Let there be $n = 100$ observations with a single predictor and a quantitative response. I fit two regression models to the data, i.e. (1) linear : $Y = \beta_0 + \beta_1 X + \epsilon$ and (2) cubic : $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon$. Suppose that the true relationship between X and Y is linear. Consider the training Residual Sum of Squares (RSS) for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, same, or is there not enough information to tell? Justify your answer.
- (b) A 99% t-based confidence interval for simple linear regression coefficient β_1 (slope) is $[3.32, 4.98]$. What is your conclusion about considering β_1 (slope) in the regression model? Justify your answer.
- ✓(c) Explain how to perform F-test for linear regression.
- ✓(d) Define GINI index and explain its use in data science.
- ✓(e) Explain the differences between Principal Component Analysis and Linear Discriminant Analysis.

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- (f) Explain type-I error and type - II errors in hypothesis testing.
- (g) Explain how the ensemble method Random Forest increases the accuracy of a classifier over a Decision Tree.
3. (a) Derive the coefficients of Multiple Linear Regression minimizing least square errors.
- (b) Describe the method to decide the accuracy of the linear regression model for best fit. 5+5
4. (a) Define Singular Value Decomposition (SVD) and state its significance in data science.
- (b) Give the steps of SVD justifying each step.
- (c) Consider a 3×2 matrix $A = \begin{bmatrix} 1,1 \\ 0,1 \\ -1,1 \end{bmatrix}$ and perform singular value decomposition of A. 3+4+3
5. Given the **time** to obtain a driving license (in years), **gender** (male, female), residential **area** (urban, rural), and the **risk** class (low, high) of a car driver, predict the risk class of a car driver by constructing a decision tree. For splitting, use information gain as a measure of impurity. 10

ID	time	gender	area	risk
1	1-2	m	urban	low
2	2-7	m	rural	high
3	>7	f	rural	low
4	1-2	f	rural	high
5	>7	m	rural	high
6	1-2	m	rural	high
7	2-7	f	urban	low
8	2-7	m	urban	low

6. (a) Are K-means and K-medoid algorithms supervised learning methods? Justify.
- (b) Differentiate between K-means and K-medoid algorithm.
- (c) Given the following 6 data points (A, B, ..., F), assuming Euclidean distance of measure, perform one iteration of K-means and K-medoid algorithm for $k = 2$. Select D and E as initial medoid and initial mean. 1+4+5

	A	B	C	D	E	F
x_1	2	7	1	8	3	9
x_2	3	2	1	5	6	7

7. (a) Is Support Vector Machine (SVM) an unsupervised algorithm?
 (b) State the objective function used by SVM for binary classification.
 (c) Derive the decision rule used in SVM to classify a given data point in binary classification. 1+3+6
8. (a) Describe the logit model in logistic regression for binary classification problem.
 (b) Given the following confusion matrix obtained for a binary classification model, compute Accuracy, Precision, Recall, F1-Score, and comment on the model.

	Prediction – Positive	Prediction – Negative	
Actual – Positive	(TP) 30	(FP) 15 FN	
Actual – Negative	(FN) 5 (FP)	(TN) 950	5+5
